Preprint

SUPPLEMENTARY MATERIAL

# Chiral Holonomy of the Closed Tetrad Frame

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**Abstract**
The eight null directions of the closed tetrad frame are the corners of a cube, four on TetA and four on TetB, and every cube edge joins the two. We transport a frame around the cube's Hamiltonian cycles in two ways: *uniformly*, with the rotation $R_{ij}$ that carries $v_i$ to $v_j$ at every step, and *chirally*, with its transpose $R_{ij}^T$ at every step leaving a TetB direction. Transposition keeps the diagonal and exchanges the two triangles. Uniform transport is parallel transport on the sphere. Every Hamiltonian cycle encloses exactly half the sphere, so the holonomy is the identity on vectors and exactly $-1$ on spinors: a full turn, undone only by a second circuit. Chiral transport never closes. Its holonomy angles are exactly $\arccos(-4639/6561)$ and $\arccos(-5983/6561)$, one for each sense of circulation, so the closure is chiral. Exchanging the labels TetA and TetB is the same as reversing the sense. Weighting each step by the shares $\mu$ and $1-\mu$ gives a rotation by $\arctan(\sigma \tan\theta)$ with $\sigma = (x-1)/(x+1)$; the chirality is odd in $\sigma$ and vanishes at $x = 1$.

**Keywords:** holonomy, parallel transport, stellated octahedron, cube, chirality, spinor, double cover, proof

## Setting

Let $v \in \{(\pm1, \pm1, \pm1)/\sqrt{3}\}$ be the eight corners of the cube on the unit sphere, the null directions of the closed frame [2, 1]. Following [1], TetA is the set with $v_1 v_2 v_3 > 0$ and TetB the set with $v_1 v_2 v_3 < 0$; parity $v \to -v$ exchanges them. (Theorem 3 shows the choice of label does not matter.) Corners $v_i, v_j$ are adjacent when they differ in one coordinate; then they lie in opposite sets and $v_i \cdot v_j = 1/3$, so the angle between them is $\theta = \arccos(1/3)$. The cube graph has six Hamiltonian cycles, twelve directed cycles from any given corner.

For adjacent $i, j$ let $n_{ij} = v_i \times v_j / |v_i \times v_j|$, $K$ its cross-product matrix, and

$$R_{ij} = I + \sin\theta\, K + (1 - \cos\theta)\, K^2, \qquad (1)$$

the rotation by $\theta$ about $n_{ij}$. For a directed cycle $C = (c_0, \ldots, c_7)$ define

$$\begin{aligned} H_u(C) &= R_{c_7 c_0} \cdots R_{c_1 c_2} R_{c_0 c_1}, \\ H_\chi(C) &= T_{c_7 c_0} \cdots T_{c_1 c_2} T_{c_0 c_1}, \end{aligned} \qquad (2)$$

with $T_{ij} = R_{ij}$ if $v_i \in$ TetA and $T_{ij} = R_{ij}^T$ if $v_i \in$ TetB. We call these the uniform and the chiral transport.

*Reading.* In the tetrad connection each $K \times K$ slice is $6 + 4 + 6$: an upper triangle, a shared diagonal and a lower triangle, $64 = 24 + 16 + 24$ in all, with TetA and TetB on the two triangles [3]. Transposition is exactly the exchange of the triangles with the diagonal kept (Lemma 1). The torsion-free connection keeps $16 + 24 = 40$ and cannot tell the two roots apart [4]; uniform transport, reading every step as TetA, is its analogue. These readings motivate the definitions; no proof below uses them.

## Two Lemmas

**Lemma 1 (the edge rotation).** *$R_{ij}$ is a rotation with $R_{ij} v_i = v_j$, and $R_{ij}^T = R(n_{ij}, -\theta) = R_{ji} = R_{ij}^{-1}$. Transposition leaves the diagonal of $R_{ij}$ unchanged and exchanges its strict upper and lower triangles.*

**Proof.** $v_i \perp n_{ij}$ and Eq. (1) turns the plane $\perp n_{ij}$ by $\theta$, from $v_i$ towards $v_j$. $K^T = -K$ and $(K^2)^T = K^2$, which give $R_{ij}^T$; and $n_{ji} = -n_{ij}$ gives $R_{ji}$. $K$ has zero diagonal and $K^2$ is symmetric. □

**Lemma 2 (turning and area).** *Along a Hamiltonian cycle drawn with great-circle arcs, every turn is $\pm\pi/3$, and every Hamiltonian cycle makes four turns of each sign. The region on its left therefore has area exactly $2\pi$.*

**Proof.** The three edges at a corner leave it at 120° to one another (by the threefold symmetry about that corner), so a path entering on one and leaving on another turns by $\pm\pi/3$. The count of four and four is checked on all six cycles. By Gauss–Bonnet on the unit sphere the region on the left has area $2\pi - \sum(\text{turns}) = 2\pi$. □

## Uniform Transport

**Theorem 1.** *For every Hamiltonian cycle, $H_u(C) = I$, while the $SU(2)$ lift of the transport, with step $U_{ij} = \cos\frac{\theta}{2}\,\mathbf{1} - i \sin\frac{\theta}{2}\, n_{ij} \cdot \boldsymbol{\sigma}$, is exactly $-\mathbf{1}$. Two circuits give $+\mathbf{1}$. Reversing the cycle transposes $H_u$.*

**Proof.** $R_{ij}$ is the Levi-Civita parallel transport of the sphere along the arc from $v_i$ to $v_j$: it is the rotation about the arc's pole, which carries the tangent plane along the arc. Around a closed geodesic polygon the transport is the rotation about the starting point by the enclosed area, here $2\pi$ (Lemma 2), hence $H_u = I$. For spin $\frac{1}{2}$ the same loop gives the factor $e^{-i\Omega/2}$ with solid angle $\Omega = 2\pi$ [6], that is $-1$. Both are confirmed by exact computation on all twelve cycles, each $U_{ij}$ covering $R_{ij}$. Reversal replaces each $R_{ij}$ by $R_{ji} = R_{ij}^T$ in the opposite order, which is $H_u^T$. □

One circuit is a full turn: vectors return, spinors return with a minus sign, and only a second circuit restores them. Uniform transport cannot tell the two senses of circulation apart, since $H_u$ and $H_u^T$ are the same rotation.

## Chiral Transport

**Theorem 2.** *For every Hamiltonian cycle, $H_\chi(C) \neq I$, and its rotation angle $\Theta$ satisfies*

$$\cos\Theta \in \left\{ -\frac{4639}{6561}, -\frac{5983}{6561} \right\}, \qquad 6561 = 3^8, \quad (3)$$

*that is $\Theta = 134.996°$ or $155.770°$. The two senses of each undirected cycle give the two values. The $SU(2)$ half-traces are $31/81$ and $-17/81$. $\Theta$ does not depend on the starting corner.*

**Proof.** Exact computation on all twelve cycles, with $\cos\theta = 1/3$, $\sin\theta = 2\sqrt{2}/3$, $\cos\frac{\theta}{2} = \sqrt{2/3}$. The $SU(2)$ values agree with Eq. (3) through $\cos\Theta = 2\cos^2\frac{\Theta}{2} - 1$: $2(31/81)^2 - 1 = -4639/6561$ and $2(17/81)^2 - 1 = -5983/6561$. Moving the starting corner conjugates $H_\chi$. □

The first angle is close to $3\pi/4$ but not equal to it: $\cos(3\pi/4) = -0.70711$ against $-4639/6561 = -0.70706$.

**Theorem 3 (relabelling is reversal).** *Let $H'_\chi$ be the chiral transport with the labels TetA and TetB exchanged. Then, for every cycle,*

$$H'_\chi(C) = H_\chi(\bar{C})^T, \quad (4)$$

*where $\bar{C}$ is $C$ traversed in the opposite sense. So exchanging the two chambers exchanges the two values of Eq. (3), exactly as reversing the sense does.*

**Proof.** Write $M_k = T_{c_k c_{k+1}}$, so $H_\chi(C) = M_7 \cdots M_0$. The step of $\bar{C}$ from $c_{k+1}$ to $c_k$ leaves the other chamber. By Lemma 1 it is $R_{c_{k+1}c_k} = R^T_{c_k c_{k+1}}$ if $c_{k+1} \in$ TetA, that is, if $c_k \in$ TetB, and $R^T_{c_{k+1}c_k} = R_{c_k c_{k+1}}$ otherwise. Either way it equals $M_k$, so $H_\chi(\bar{C}) = M_0 M_1 \cdots M_7$. Relabelling transposes every step, so $H'_\chi(C) = M_7^T \cdots M_0^T = (M_0 \cdots M_7)^T$. □

The two senses use the same eight steps in opposite orders. For uniform transport the orders give transposes, the same rotation; for chiral transport they give different rotations. That difference is the chirality.

## Weighting by the Shares

Weight each step by the shares $\mu$ and $1-\mu$ of the constitutive law, $\mu(x) = x/(x+1)$ [5]: $S_{ij} = \mu R_{ij} + (1-\mu)R_{ij}^T$ leaving TetA and $\mu R_{ij}^T + (1-\mu)R_{ij}$ leaving TetB. Let $H_\sigma(C)$ be the product, with $\sigma = 2\mu - 1$.

**Proposition 4.** *(a) $S_{ij} = I + \sigma\sin\theta\, K + (1-\cos\theta)K^2$ leaving TetA. It is normal, with eigenvalue 1 along $n_{ij}$ and $\cos\theta \pm i\sigma\sin\theta$ in the plane: a rotation by $\varphi$, with*

$$\tan\varphi = \sigma\tan\theta, \quad (5)$$

*times a contraction by $\rho = (\cos^2\theta + \sigma^2\sin^2\theta)^{1/2} \geq 1/3$. (b) $\sigma = (x-1)/(x+1)$, the difference of the shares, and $x \to 1/x$ sends $\sigma \to -\sigma$. (c) $H_{-\sigma}(C) = H_\sigma(\bar{C})^T$. (d) At $\sigma = 0$ ($x = 1$) every step is symmetric and the two senses have the same rotational part; the difference between the senses is odd in $\sigma$. $\sigma = \pm 1$ give Theorem 2.*

**Proof.** (a) Substitute Eq. (1); $S_{ij}$ is a polynomial in $K$, hence normal, and the plane block is $\begin{pmatrix} \cos\theta & -\sigma\sin\theta \\ \sigma\sin\theta & \cos\theta \end{pmatrix}$. (b) $2x/(x+1) - 1$. (c) Leaving TetB the step at $\sigma$ is the TetA step at $-\sigma$, its transpose; the argument of Theorem 3 applies. (d) At $\sigma = 0$ the step is $I + (1-\cos\theta)K^2$, symmetric, and by (c) $H_0(C) = H_0(\bar{C})^T$; a matrix and its transpose have transposed polar factors, hence equal rotation angles. By (c) the difference changes sign with $\sigma$. □

## Summary

On the cube of null directions, uniform transport is parallel transport on the sphere. Each Hamiltonian cycle encloses half the sphere, so it returns vectors unchanged and spinors with a minus sign. Chiral transport, with the TetB steps transposed, never closes. Its two exact angles belong to the two senses of circulation, and exchanging the chambers exchanges them. Weighted by the shares of the constitutive law, each step rotates by $\arctan(\sigma\tan\theta)$, and the chirality is odd in $\sigma = (x-1)/(x+1)$ and vanishes at $x = 1$.

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